

Contextual Intelligence Engine (CIE)

Operational Context Modeling & Feature Taxonomy Report

OVERVIEW & SCOPE

This report models operational context for the Cyber Threat Intelligence (CTI) Platform using the official TON-IoT Processed Network Dataset (Network_dataset_*.csv).

Every feature in the 45-feature network schema is categorized into exactly one of 7 context groups:

1. Device Context (4 features): Source/Destination IP addresses and X.509 SSL certificate identity.
2. Network Context (5 features): Transport/Service protocols, socket ports, and TCP handshake state
3. Traffic Context (9 features): Flow durations, byte throughput, packet counts, and payload length
4. Temporal Context (1 feature): Unix epoch timestamp for chronological event ordering.
5. Security Context (11 features): Zeek anomaly flags, SSL protocol strength, and ground-truth targ
6. Behavioral Context (14 features): Application payload interactions across HTTP and DNS protocols
7. Environmental Context (1 feature): Client browser/OS User-Agent software environment.

SUMMARY DISTRIBUTION TABLE

Context Category	Feature Count	Share (%)	Key CTI Contribution
Behavioral Context	14 features	31.1%	HTTP & DNS payload interaction behavior
Security Context	11 features	24.4%	Protocol anomalies & ground-truth labels
Traffic Context	9 features	20.0%	Flow duration & packet/byte throughput
Network Context	5 features	11.1%	Transport protocols & connection states
Device Context	4 features	8.9%	IP endpoint & certificate identities
Temporal Context	1 feature	2.2%	Event timestamp & inter-arrival time
Environmental Context	1 feature	2.2%	Client User-Agent & OS environment
TOTAL	45 features	100.0%	Unified Multi-Layer Operational Context

Context Feature Mapping Table (Part 1: Device, Network, Traffic)

Feature	Context Category	Nature	Score	CTI Contribution
src_ip	Device Context	Static (per flow)	9.0	Direct: Essential for origin host t...
dst_ip	Device Context	Static (per flow)	9.0	Direct: Critical for identifying ta...
ssl_subject	Device Context	Static (per cert)	8.0	Indirect: Helps verify server ident...
ssl_issuer	Device Context	Static (per cert)	7.5	Indirect: Useful for detecting self...
src_port	Network Context	Dynamic	7.5	Indirect: Helps track socket reuse ...
dst_port	Network Context	Dynamic	8.5	Direct: Essential for pinpointing t...
proto	Network Context	Static (per flow)	9.5	Direct: Governs stateful vs. statel...
service	Network Context	Dynamic	9.0	Direct: Crucial for applying applic...
conn_state	Network Context	Dynamic	9.5	Direct: High indicator for TCP SYN ...
duration	Traffic Context	Dynamic	8.5	Direct: Distinguishes persistent ba...
src_bytes	Traffic Context	Dynamic	9.0	Direct: Key metric for detecting da...
dst_bytes	Traffic Context	Dynamic	9.0	Direct: Critical for identifying la...
src_pkts	Traffic Context	Dynamic	8.5	Direct: Essential metric for identi...
dst_pkts	Traffic Context	Dynamic	8.5	Direct: Differentiates interactive ...
src_ip_bytes	Traffic Context	Dynamic	8.0	Indirect: Provides exact network la...
dst_ip_bytes	Traffic Context	Dynamic	8.0	Indirect: Evaluates total network b...
http_request_body_len	Traffic Context	Dynamic	8.0	Direct: Detects large HTTP payload ...
http_response_body_len	Traffic Context	Dynamic	8.0	Direct: Identifies web page leakage...
ts	Temporal Context	Dynamic	8.5	Direct: Crucial for event correlati...
weird_name	Security Context	Dynamic	9.5	Direct: Immediate indicator of malf...

Context Feature Mapping Table (Part 2: Security, Behavioral, Environmental)

Feature	Context Category	Nature	Score	CTI Contribution
weird_notice	Security Context	Dynamic	9.0	Direct: High-fidelity security alert...
weird_addl	Security Context	Dynamic	8.0	Indirect: Supplies additional diagn...
missed_bytes	Security Context	Dynamic	7.0	Indirect: Highlights network interf...
dns_rejected	Security Context	Dynamic	8.0	Direct: Indicates DNS firewall bloc...
ssl_established	Security Context	Dynamic	8.5	Direct: Failed SSL handshakes indic...
ssl_version	Security Context	Dynamic	8.5	Direct: Identifies outdated, vulner...
ssl_cipher	Security Context	Dynamic	8.5	Direct: Detects weak or deprecated ...
ssl_resumed	Security Context	Dynamic	6.5	Indirect: Contextualizes session ti...
label	Security Context	Static (Target)	10.0	Direct: Core supervised target for ...
type	Security Context	Static (Target)	10.0	Direct: Core target for multi-class...
dns_query	Behavioral Context	Dynamic	8.5	Direct: Detects DGA domains, C2 ser...
dns_qclass	Behavioral Context	Dynamic	6.0	Indirect: Verifies standard DNS que...
dns_qtype	Behavioral Context	Dynamic	7.0	Direct: Detects DNS tunneling via u...
dns_rcode	Behavioral Context	Dynamic	8.0	Direct: High NXDOMAIN rates indicat...
dns_AA	Behavioral Context	Dynamic	6.5	Indirect: Helps detect DNS spoofing...
dns_RD	Behavioral Context	Dynamic	6.5	Indirect: Contextualizes recursive ...
dns_RA	Behavioral Context	Dynamic	6.5	Indirect: Evaluates recursive DNS a...
http_trans_depth	Behavioral Context	Dynamic	7.0	Indirect: Detects HTTP session pipe...
http_method	Behavioral Context	Dynamic	9.0	Direct: Critical for identifying we...
http_uri	Behavioral Context	Dynamic	9.5	Direct: Primary feature for detecti...
http_version	Behavioral Context	Dynamic	6.5	Indirect: Detects anomalous or anci...
http_status_code	Behavioral Context	Dynamic	8.5	Direct: High 404 rates signal direc...
http_orig_mime_types	Behavioral Context	Dynamic	7.5	Direct: Identifies malicious file u...
http_resp_mime_types	Behavioral Context	Dynamic	7.5	Direct: Identifies returned payload...
http_user_agent	Environmental Context	Static (per client)	8.5	Direct: Key feature for identifying...